q3_bitwise.c - Code::Blocks / VS Code (C)

```c
#include <stdio.h>

enum { LIGHT = 0, FAN = 1, AC = 2, HEATER = 3, PUMP = 4 };

unsigned char switch_on(unsigned char reg, int dev)  { return reg | (1u << dev); }
unsigned char switch_off(unsigned char reg, int dev) { return reg & ~(1u << dev); }
unsigned char toggle(unsigned char reg, int dev)     { return reg ^ (1u << dev); }
int is_on(unsigned char reg, int dev)                { return (reg >> dev) & 1u; }

void show(unsigned char reg)
{
    for (int bit = 7; bit >= 0; bit--) putchar(((reg >> bit) & 1u) ? '1' : '0');
    putchar('\n');
}

int count_on(unsigned char reg)
{
    int c = 0;
    while (reg) { reg &= reg - 1; c++; }
    return c;
}

int main(void)
{
    unsigned char reg = 0;
    reg = switch_on(reg, LIGHT);
    reg = switch_on(reg, AC);
    show(reg);
    reg = toggle(reg, FAN);
    reg = switch_off(reg, LIGHT);
    show(reg);
    printf("AC is %s, PUMP is %s\n", is_on(reg, AC) ? "ON" : "OFF",
           is_on(reg, PUMP) ? "ON" : "OFF");
    printf("Devices ON = %d\n", count_on(reg));

    unsigned char r = 0x06; int dev = 2;
    printf("reg & 1 << dev   = %d\n", r & 1 << dev);
    printf("(reg & 1) << dev = %d\n", (r & 1) << dev);

    signed char sc = (signed char)0x80;
    printf("signed 0x80 >> 7 = %d\n", sc >> 7);
    unsigned char uc = 0x80;
    printf("unsigned 0x80 >> 7 = %d\n", uc >> 7);
    return 0;
}
```

Terminal - output

```
user@pc:~/Assignment2$ gcc -Wall q3_bitwise.c -o q3 && ./q3
00000101
00000110
AC is ON, PUMP is OFF
Devices ON = 2
reg & 1 << dev   = 4
(reg & 1) << dev = 0
signed 0x80 >> 7 = -1
unsigned 0x80 >> 7 = 1
user@pc:~/Assignment2$ _
```